Jai Sheng Ong

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Education

University of Cincinnati Cincinnati, OH
Bachelor of Science, Computer Science — GPA: 3.57 *Expected May 2028*

- Relevant Coursework: Data Structures, Database Design & Development, Software Engineering

Cincinnati State Technical and Community College Cincinnati, OH
Associate of Applied Science, Software Engineering Technology — GPA: 3.74 *Graduated: Spring 2024*

- Relevant Coursework: System Analysis & Design, Python Object-Oriented Programming

Experience

Software Developer T&I Aug 2026 – Present
Siemens Digital Industries Software | Internship *United States*

- Conducting Research and Development on innovative algorithms to confront business needs.

Machine Learning Researcher Jan 2026 – May 2026
Ming Chi University of Technology | Internship *Taiwan*

- **Architected an end-to-end machine learning pipeline** in Python to predict highly volatile industrial wastewater metrics.
- **Engineered advanced time-series features** by implementing a **2D-state Kalman filter** and **ARIMA** one-step-ahead forecasting, effectively de-noising sensor volatility and completely eliminating temporal data leakage.

Software Engineer May 2025 – Aug 2025
Aubot | Internship *Remote*

- **Reduced lesson publishing time by 30%** by auditing and correcting 120+ technical, grammar, and clarity errors across 10+ advanced **Python** course documents.
- **Strengthened content reliability** by authoring automated test scripts to catch incorrect solution methods and contributing targeted fixes via **Git** pull requests across a collaborative codebase.

Projects

WasteWater-COD-Forecast | *Python, PyCaret, Jupyter, Optuna*

- Preprocessed a small-scale industrial sensor dataset from MCUT's wastewater facility by studying plant process context, mapping raw feature semantics to physical treatment stages, and resolving missing values.
- Applied VIF-based feature selection to remove multicollinear predictors, motivating time-series feature engineering via **Kalman filtering** and **ARIMA** forecasting to surface temporal patterns.
- Trained an **Extra Trees** regressor on the engineered features and analyzed importance scores to interpret which derived signals most strongly drove target predictions.
- Implemented **LSTM** as a sequence-learning baseline; attributed its underperformance to insufficient training samples for deep learning, empirically validating Extra Trees as the appropriate model for small-scale edge deployment.

webMessenger | *Node.js, Express, Socket.io, MongoDB Atlas, Azure*

- Recreated a real-time web messenger (Node.js, Socket.io, MongoDB Atlas) with a **5-person Agile team** across 2 sprints, delivering **14 features** including authenticated chat, private messaging, and real-time presence tracking.
- Owned **Azure App Services** deployment and **GitHub Actions CI/CD pipelines** across 3 branches; integrated the **MongoDB** database layer connecting the full server and storage stack for the team's Sprint 2 features.
- Applied **defense-in-depth security** across all layers — **bcrypt** password hashing, CSP headers, and **DOMPurify** XSS sanitization to protect against XSS, NoSQL injection, and credential attacks

Technical Skills

Languages: Python, C++, VB, Assembly, MatLab, NI LabVIEW, R
Web Development: JavaScript, HTML, CSS, PHP
Frameworks & Libraries: React, Node.js, Express, Socket.io, PyCaret, Optuna
Databases: MySQL, MongoDB
Tools & Platforms: Git, GitHub, Azure, Linux (Ubuntu), LaTeX, Jupyter Notebook
Product Skills: Agile Methodologies, Scrum, Data Analysis (SQL, Python, Excel)

Available for Co-op/Internship: **Summer 2027**